An Children Autism Spectrum Disorder Diagnosis using Convolutional Neural Network and Recurrent Neural Network

Yingtong Ai
Preschool and Special Education College
Kunming University
Kunming, China
ayt.baby@126.com

2024 International Conference on Distributed Systems, Computer Networks and Cybersecurity (ICDSCNC) | 979-8-3503-7544-2/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICDSCNC62492.2024.10941426

***Abstract*—In recent times, Autism is neuro-developmental-disorder categorized by discrepancies in social communication, verbal and non-verbal communication accompanied by partial interests as well as monotonous stereotypical behaviours. The traditional models which Assisted diagnosis of childhood autism i.e. Autism-Spectrum-Disorder (ASD) failed in classification framework of the model and fell short of detecting the nuanced heterogeneity of ASD. Therefore, this paper proposes Convolutional Neural Network- Recurrent Neural Network (CNN-RNN) for ASD detection is employed on Autism Brain Imaging Data Exchange (ABIDE) dataset which incorporates Functional Magnetic Resonance Imaging (fMRI) images as well as effectively detected complex ASD and Non-Autism Spectrum Disorder (Non-ASD) images. Primarily, the required and most relevant features are extracted using CNN by max pooling layers convolutional layers, and dense layers. Then, the dimensionality of the extracted features are reduced by using Pearson Correlation Coefficient (PCC). Ultimately, these features are classified using RNN from fMRI scan images into ASD and Non-ASD images. These images helped in quick identification of children with disorder and children without disorder. The proposed CNN-RNN attained better results in respect of accuracy 99.50%, recall 96.21%, precision 95.23% and f1-score 94.67% when compared with existing Weisfeiler Lehman- Deep Graph Convolutional Networks (WL-Deep GCN).**

Keywords—autism spectrum disorder, autism brain imaging data exchange, functional magnetic resonance imaging, non-autism spectrum disorder, pearson correlation coefficient.

I. Introduction

In recent times, around 1.5% of the population is exaggerated by Autism Spectrum Disorder (ASD) which is frequently diagnosed in initial childhood with lifelong neurodevelopmental conditions, by affecting their social communication and interactions [1]. These children with ASD faced deficiency of eye contact and remained unresponsive when anyone called by their names due to their symptoms which affects children's mental state, [2]. The Detection of ASD in adulthood is difficult for clinicians to gain information about the developmental history of affected persons and ASD represents a spectrum of experiences and abilities where every person has their unique traits to contribute their way in navigating their life and connecting with others [3]. The diagnosis of ASD is not done until 2-3 years where these symptoms appeared in childhood but it is done at the age of 4 years and detection of autism is difficult because it needs long time to improve case [4]. Moreover, adults with high intelligence developed coping and camouflaging strategies where these are present from their childhood which makes difficult for parents to capture underlying difficulties among them [5]. The classification of ASD images using FMRI images helped in early detection of autism disorder and detection at younger age enables the timely intervention which improves outcomes for detection of ASD [6]. State of art methods includes CNN for ASD detection which identified distinct brain structures and characterized complex physiocratic disorder but, accuracy is slightly slower than latest version of ABIDE dataset [7]. Simultaneously, Long Short-Term Memory (LSTM) for Assisted diagnosis of childhood autism detection effectively recognized ASD children's action and original skeleton data is lost because of overlapping and occlusion [8]. 3D CNN has utilized for visualization of combination brain regions which classified the categorization of brain sections proficiently for ASD detection [9]. Additionally, different transfer learning methods like VGG19 and Resnet50 classified facial images of children for detecting the ASD among autism images and non-autism images [10].

The main contributions of this research include -

- The CNN used for feature extraction which automatically identified relevant features by convolutional, max pooling and dense layers.
- The proposed CNN-RNN for ASD detection is pre-processed by PCC to reduce dimensionality and identify highly correlated variables.
- The proposed CNN-RNN effectively detected ASD into spectrum disorder and typical images which are measured in respect of accuracy, recall, precision and f1-score.

This research is organized as following: related work of ASD is discussed in section 2, and the overall methodology is explained in section 3, section 4 evaluates experimental results and results are discussed in section 5 and finally this paper ends with conclusion in section 6.

II. Literature Review

Juncai Xu et al. [11] suggested an advanced Convolutional Neural Network (CNN) algorithm to inspect the functional Magnetic Resonance Imaging (fMRI) images towards recognition of Autism-Spectrum Disorder (ASD). The suggested CNN integrated pooling and fully connected layers which provided optimization for data interpretation of high dimensional values. The 4D image provided by fMRI had converted to 2D image and given as input to CNN for better data visualization of images. The detection of ASD achieved more accuracy by using the proposed CNN method. However, the suggested CNN model is not useful for real-world clinical applications and decision making, specifically for better screening and treatment of autism.

979-8-3503-7544-2/24/$31.00 ©2024 IEEE

Authorized licensed use limited to: BTU Cottbus-Senftenberg. Downloaded on November 08,2025 at 10:11:20 UTC from IEEE Xplore. Restrictions apply.

David Abramian et al. [12] suggested augmented methods for Assisted diagnosis of childhood autism using 3D CNN for fMRI identification in small children. The 3D CNN was accomplished using keras platform and comprised 3 neural layers' dense layer, max-pool layer, and final one-node layer. Adam optimizer had been used for training of CNN to prevent overfitting till the accuracy improved of ASD detection. The classification accuracy increased by using the proposed 3D CNN model which helped in simple detection of ASD. However, 3D-CNN needed a greater number of computational resources to extract features because of the large size of medical images.

Hidir Selcuk Nogay et al. [13] presented a CNN method for classification of ASD through considering age and gender of children. The suggested CNN model developed using Glowworm Swarm Optimization (GSO) algorithm and tested with five validation techniques. These five techniques were divided into 3 models: one model for age, one model for gender and other model for both age and gender classification. The proposed CNN model achieved more effectiveness in detection of ASD using age and gender factors. However, CNN model caused overfitting due to relatively availability of less data.

Kambham sravani et al. [14] suggested Canny Deriche Edge Detection Deep Convolutional Neural Networks Dipper-Throated Particle Swarm Optimization CDED-DCNN-DTPSO for identification of ASD. Initially, in preprocessing stage image edges are identified and unclear images rejected using CDED. DTPSO algorithm has used for tuning the DCNN and Class Activation Mapping (Grad-CAM) for picturing the decision-making procedure. This method attained more accuracy in detecting ASD and in classifying the ASD and Non-ASD. However, this CDED-DCNN-DTPSO model required a greater amount of data and a feature set for training as well and model had less robust on extracted data.

Mingzhi Wang et al. [15] suggested a Weisfeiler Lehman-Deep Graph Convolutional Networks (WL-Deep GCN) for ASD diagnosis. Initially, WL-Deep GCN model extracted the images from fMRI scan, Weight learning network is used for representing comparability between imaging and non-imaging data. Then, Edge Drop strategy utilized for removing edges in graphs and reduces the overfitting problem. The WL-Deep GCN achieved substantial performance in diagnosis of ASD identification. However, this WL-Deep GCN incapable to acknowledge brain aberration and contrast among patients.

Hasan Alkahtani et al. [16] demonstrated CNN methods for identification of ASD in small children. CNN utilized transfer learning models like MobileNetV2 and hybrid VGG19 with deep learning models for feature extraction of autistic and non-autistic images of children. These images had classified using machine learning models like random forest, Linear SVC, decision tree, MLP Classifier gradient boosting, and K-nearest neighbour methods. The proposed CNN addressed a highly accurate detection of ASD in the field of computer vision. However, CNN not provided accurate results when screening was performed for individual children.

Amna Hendr et al. [17] established autism spectrum disorder based on CNN using DL techniques. The suggested CNN developed a handwritten dataset for ASD and non-ASD people that is based on drawing and writing. The model aimed for planning a resource of low demand computer-based diagnosis of ASD disease which was implemented on both health and education domains for enabling reliable management of persons undergoing ASD. The CNN exhibited impressive performance in identifying the early ASD by hand written task. However, handwritten dataset applicable for only children with mild ASD and not applicable to children with severe ASD.

III. Methodology

In this research, CNN-RNN is proposed for ASD detection and Assisted diagnosis of childhood autism that is employed on publicly available i.e. ABIDE dataset. Initially, the ABIDE dataset consists of fMRI images then, specific features ae extracted using CNN. Additionally, for dimensionality reduction PCC is utilized. After that, from pre-processed images features are classified into ASD and Non-ASD images by using RNN. The overall methodology is shown in Figure 1.

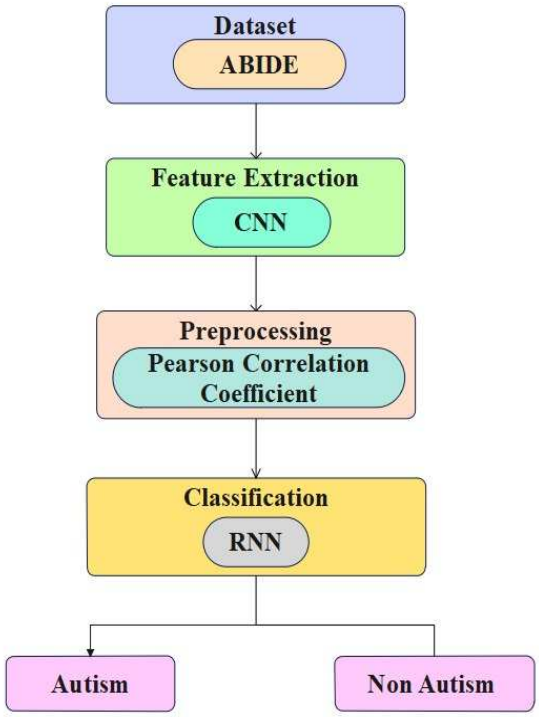

Fig. 1. Overall block diagram of proposed methodology

A. Dataset

The proposed CNN-RNN is employed on ABIDE dataset [18], which is publicly available in Kaggle this dataset consists of 6214 files and 32 directories of images related to ASD. These files provide FMRI scan images of small children and the participants in these data are 7–64 years old, including people with ASD and without ASD. The features in this data are extracted by using CNN to identify the most appropriate and relevant features which is explained below.

B. Feature Extraction

ABIDE dataset is utilized by CNN to identify and extract relevant features. The obtained brain images from ABIDE data are extracted into various layers like max pooling layers' dense layers and convolutional layers which are converted from complex images into simple images. Below Figure 2 explains about feature extraction which is accomplished using convolutional layers, max pooling layers and dense layers.

Authorized licensed use limited to: BTU Cottbus-Senftenberg. Downloaded on November 08,2025 at 10:11:20 UTC from IEEE Xplore. Restrictions apply.

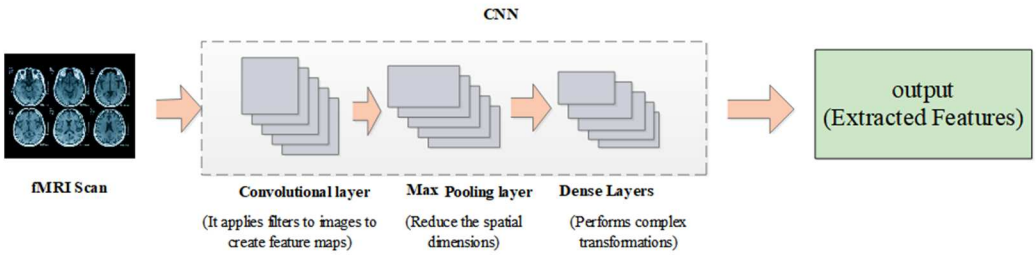

Fig. 2. Feature extraction using CNN

1) Convolutional layers: Convolutional layers are a crucial element for CNN, it incorporates a set of kernels which are learned during the training. These kernels are involved with every fMRI image and create activation maps for further processes. For convolution the kernel glides among height and width of image and at every single spatial position dot product between filter and input has calculated. By repeating this procedure for every fMRI image finally, an activation map is generated this map is assembled for every filter along with depth dimension for generating output volume. Convolutional layers are a crucial element for CNN, it extracts features from abide dataset that includes images by imposing numerous filters to recognise patterns like shape, size, gap etc. additionally, these layers transmit the fMRI mages into feature maps by priorly recognising sophisticated features and deep neural network layers assists by detecting complex patterns. This output neuron is related to every component of the activation map, which indicates these neurons are connected to all small regions. These connections between layers provide better sharing of neuron information in activation maps and parameters with each other, this increases the maximum response of neurons for every local region input. The calculation of the operation described in equation (1)

$$O_{ij} = \sum_m \sum_n D_{(i+m)(j+n)} \times K_{mn} \tag{1}$$

O_{ij} is feature map output, K means kernel, D is image input, and n, m are proportions of k.

2) Max pooling layers: This layer reduces the size of an fMRI image by dividing it into rectangular regions and calculating the maximum value of each region. The activation map created by the convolutional layer is further processed by max pooling layers. These layers employed on convolutional layers of CNN max pooling layers also include slider filters similar to convolutional layers but it obtained maximum values within filters instead of performing matrix multiplication. These layers reduce the space-based dimension volume of input, and highlight conspicuous features better than an effective analytical process. The equation for these layers given as represented in equation (2).

$$M = Maximum(R) \tag{2}$$

Here M is maximum pooled output and R represents the region.

3) Dense layer: A dense layer is a layer of neurons in a CNN that receives input from all the neurons in the previous layer [19]. These layers learn from the combination of features across the entire image and can model more complex, higher-level patterns. It also assists in predicting the severity of ASD and also facilitates in diagnosing ASD efficiently. The weights between neurons are trained through backpropagation, adjusting to minimize the error in the model's predictions. This prediction is the final prediction of the classification process that provides final outputs of the network. The following equation (3) defines the mathematical expression of dense layers.

$$y = Activation(W \times x + b) \tag{3}$$

Where y represents output, W are weights, x is input, and b is bias term. The output from dense layers have taken as input to the preprocessing which uses PCC for preprocessing technique.

C. Preprocessing

The input from the CNN is pre-processed by using Pearson Correlation Coefficient (PCC) [20]. The PCC is calculated by correlation between different Regions Of Interest (ROI), to provide functional connectivity matrix and reduce dimensionality and this connectivity matrix helps to categorize ASD and Non-ASD into groups. The grouping of these brain regions is done by below shown equation (4).

$$PCC(u,v) = \frac{E(uv) - E(u)E(v)}{\sqrt{E(u)^2 - E^2(u)}\sqrt{E(v^2) - E^2(v)}} \tag{4}$$

Where, $PCC\ (u,v)$ stands for correlation between time series of two brain regions u and v. E is mathematical expectation and the coefficient range is -1 to 1. Where 1 represents positive correlation and -1 is negative correlation among two brain regions. This preprocessed data is further classified using RNN.

D. Classification

The extracted features are classified using RNN for complex ASD and Non-ASD images [21]. RNN incorporates multiple fixed activation function units one for each time step, each unit has a hidden state of unit. This hidden state is modernized each time for signifying the modification in past information updated using following recurrence relation.

The formula for calculating the current state is represented in equation (5).

$$h_t = f(h_{t-1}, x_t) \tag{5}$$

Authorized licensed use limited to: BTU Cottbus-Senftenberg. Downloaded on November 08,2025 at 10:11:20 UTC from IEEE Xplore. Restrictions apply.

Where, h_t signifies current state, h_{t-1} denotes previous state and x_t denotes input state

Formula for activation function is mentioned in equation (6).

$$h_t = tanh(w_{hh}h_{t-1} + w_{xh}x_t) \quad (6)$$

Where, w_{hh} denotes recurrent neuron weight and w_{xh} denotes input neuron weight as mentioned in equation (7)

$$y_t = yW_{hy}h_t \quad (7)$$

Where, y_t represents output and yW_{hy} represents output layer.

By utilizing the above equations hierarchical labels of images are generated, these labels organised in coarse to fine pattern. Each image is assigned hierarchical labels with coarse labels categorizing them as either ASD or Non-ASD. Fine labels sub categorized the images like mild ASD, moderate ASD and severe ASD. In this way the RNN method has been utilized for classifying the images using hierarchical labels.

IV. Experimental Results

The proposed CNN-RNN diagnosis for ASD is implemented on the NVIDIA A100 Graphics unit, having fifty periods through batch size 64, Windows 10 Intel Core i7 processor with 16GB. The performance of proposed CNN-RNN is evaluated in terms of accuracy, precision, recall and F1 score shown below eq. (8) - (11).

$$Accuracy = \frac{TP+TN}{TP+FP+TN+FN} \quad (8)$$

$$Precision = \frac{TP}{TP+FP} \quad (9)$$

$$Recall = \frac{TP}{TP+FN} \quad (10)$$

$$F1\ Score = \frac{2TP}{2TP+FP+FN} \quad (11)$$

Where TN and TP signifies true negative and true positive, FN, FP refers to false negative and false positive.

A. Performance Analysis for Classification Model

The suggested CNN-RNN has trained on ABIDE dataset for ASD classification. The performance evaluation is done by Accuracy, Precision, Recall and F1 score. The performance of proposed model is associated with existing methods such as Recurrent Neural Network (RNN), Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM) which is described in below Table 1.

TABLE I. Performance Analysis of Proposed CNN-RNN with Existing Methods

Performance models	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
RNN	80.91	90.25	90.50	90.75
CNN	91.34	88.79	87.25	88.50
LSTM	92.67	89.90	95.25	95.90
Proposed CNN-RNN	99.50	95.23	96.21	94.67

From Table 1, the proposed CNN-RNN achieved better results in terms of accuracy - (99.50%), Precision - (95.23%), Recall - (96.21%), and F1 score - (94.67%) when compared with existing models such as RNN, CNN and LSTM.

B. Comparative Analysis

The Proposed CNN-RNN classifier model is compared with Convolutional Neural Network (CNN) [11], Canny Deriche Edge Detection Deep Convolutional Neural Networks Dipper-Throated Particle Swarm Optimization algorithm (CDED-DCNN-DTPSO) [14] and Weisfeiler Lehman- Deep Graph Convolutional Networks (WL-Deep GCN) [15] as represented in the Table 2.

TABLE II. Comparative Analysis

Performance models	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
CNN [11]	99.39	99.85	98.80	99.32
CDED-DCNN-DTPSO [14]	95.90	96.50	95.90	96.20
WL-Deep GCN [15]	77.27	77.70	80.96	NA
Proposed CNN-RNN	99.50	95.23	96.21	94.67

From the Table 2, Proposed CNN-SVM classification model provided better results which includes accuracy - 99.50%, Recall - 96.21%, precision- 95.23% and F1 Score - 94.67% when compared with other existing methods like CNN [11] accuracy - 99.39%, Recall - 98.80%, precision - 99.85% and F1 Score - 99.32%, CDED-DCNN-DTPSO [14] accuracy 95.90%, Recall - 95.90%, precision - 96.50% and F1 Score - 96.20%, WL-Deep GCN [15] accuracy -77.27%, Recall - 80.96%, precision - 77.70%.

V. Discussion

In this study, CNN-RNN is proposed for an Autism-spectrum disorder (ASD) for Assisted diagnosis of childhood autism which is employed on ABIDE dataset that provides fMRI scan images of the brain. Then, this data features are extracted using the CNN, CNN used three layers for extraction of features that are max pooling layers, convolutional layers and dense layers. Since various features are extracted, dimensionality reduction is imposed using PCC to process based on the specific features and eliminate the unwanted features. After that, ASD is classified using RNN which effectively classified complex images because RNN manages pixel data more effectively. The proposed CNN-RNN for ASD gave accurate results in terms of accuracy - 99.50%, Recall - 96.21%, precision 95.23% and F1 Score - 94.67% when compared with other existing methods like CNN [9] accuracy 99.39%, Recall 98.80%, precision - 99.85% and F1 Score - 99.32%, CDED-DCNN-DTPSO [12] accuracy 95.90%, Recall 95.90%, precision 96.50% and F1 Score 96.20%, WL-Deep GCN [13] accuracy 77.27%, Recall 80.96%, precision 77.70%.

VI. Conclusion

In this research, CNN-RNN proposed for ASD detection and Assisted diagnosis of childhood autism which is employed on ABIDE dataset, which contains fMRI scan images of the brain. From these image data, features are extracted by CNN to automatically identify relevant features from data using multiple layers for feature extraction. These features are further processed by using PCC by reducing dimensionality and to identify highly correlated variables. These features are classified by using RNN and effectively classified into complex ASD and Non-ASD images. These

Authorized licensed use limited to: BTU Cottbus-Senftenberg. Downloaded on November 08,2025 at 10:11:20 UTC from IEEE Xplore. Restrictions apply.

images helped in quick identification of children with disorder and children without disorder. The proposed CNN-RNN resulted in higher accuracy of 99.50% when compared with other existing deep learning models such as CDED-DCNN-DTPSO [14], WL-Deep GCN [15]. In future, hybrid models will be utilized to develop effective detection for improving more accuracy.

ACKNOWLEDGEMENT

Study on the establishment of early health education mechanism for 0-6 year old children in rural areas of Yunnan Province (ZX20230279).

REFERENCES

[1] Yin, Wutao, Sakib Mostafa, and Fang-Xiang Wu. "Diagnosis of autism spectrum disorder based on functional brain networks with deep learning." Journal of Computational Biology 28, no. 2 (2021) : 146 - 165.

[2] Ashraf, Adnan, Qingjie Zhao, Waqas Haider Bangyal, and Muddesar Iqbal. "Analysis of brain imaging data for the detection of early age autism spectrum disorder using transfer learning approaches for Internet of Things." IEEE Transactions on Consumer Electronics (2023).

[3] Liloia, Donato, Denisa Adina Zamfira, Masaru Tanaka, Jordi Manuello, Annachiara Crocetta, Roberto Keller, Mauro Cozzolino, Sergio Duca, Franco Cauda, and Tommaso Costa. "Disentangling the role of gray matter volume and concentration in autism spectrum disorder: A meta-analytic investigation of 25 years of voxel-based morphometry research." Neuroscience & Biobehavioral Reviews (2024): 105791.

[4] Mujeeb Rahman, K. K., and M. Monica Subashini. "A deep neural network-based model for screening autism spectrum disorder using the quantitative checklist for autism in toddlers (QCHAT)." Journal of Autism and Developmental Disorders 52, no. 6 (2022): 2732 - 2746.

[5] Yan, Z., Yang, H., Su, Q. and Ramesh, G.P., 2023, November. On the Development of Big Data Intelligent Module Vocabulary System. In International Conference on Cognitive based Information Processing and Applications (pp. 359-371). Singapore: Springer Nature Singapore.

[6] Rabbi, Md Fazle, SM Mahedy Hasan, Arifa Islam Champa, and Md Asif Zaman. "A convolutional neural network model for early-stage detection of autism spectrum disorder." In 2021 international conference on information and communication technology for sustainable development (icict4sd), pp. 110 - 114. IEEE, 2022.

[7] Ahmed, Ibrahim Abdulrab, Ebrahim Mohammed Senan, Taha H. Rassem, Mohammed AH Ali, Hamzeh Salameh Ahmad Shatnawi, Salwa Mutahar Alwazer, and Mohammed Alshahrani. "Eye tracking-based diagnosis and early detection of autism spectrum disorder using machine learning and deep learning techniques." Electronics 11, no. 4 (2022): 530.

[8] Zhang, Y., Tian, Y., Wu, P. and Chen, D., 2022. Application of skeleton data and long short-term memory in action recognition of children with autism spectrum disorder. Sensors, 21(2), p.411.

[9] Ke, Fengkai, Seungjin Choi, Young Ho Kang, Keun-Ah Cheon, and Sang Wan Lee. "Exploring the structural and strategic bases of autism spectrum disorders with deep learning." Ieee Access 8 (2020) : 153341 - 153352.

[10] Feng, Min, and Juncai Xu. "Detection of ASD children through deep-learning application of fMRI." Children 10, no. 10 (2023) : 1654.

[11] Kalaiselvi, A., S. Nagarathinam, Timothy Dhayakar Paul, and M. Alagumeenaakshi. "Detection of autism spectrum disorder using transfer learning." Turkish Journal of Physiotherapy and Rehabilitation 32, no. 2 (2021): 926 -933.

[12] Jönemo, Johan, David Abramian, and Anders Eklund. "Evaluation of augmentation methods in classifying autism spectrum disorders from fMRI data with 3D convolutional neural networks." Diagnostics 13, no. 17 (2023) : 2773.

[13] Nogay, Hidir Selcuk, and Hojjat Adeli. "Multiple classification of brain MRI autism spectrum disorder by age and gender using deep learning." Journal of Medical Systems 48, no. 1 (2024): 15.

[14] Sravani, Kambham, and P. Kuppusamy. "Optimized Deep Convolutional Neural Network for Autism Spectrum Disorder Detection Using Structural MRI and DTPSO." IEEE Access (2024).

[15] Wang, Mingzhi, Jifeng Guo, Yongjie Wang, Ming Yu, and Jingtan Guo. "Multimodal autism spectrum disorder diagnosis method based on DeepGCN." IEEE Transactions on Neural Systems and Rehabilitation Engineering (2023).

[16] Alkahtani, H., Aldhyani, T.H. and Alzahrani, M.Y., 2023. Deep learning algorithms to identify autism spectrum disorder in children-based facial landmarks. Applied Sciences, 13(8), p.4855.

[17] Liu, X., Hasan, M.R., Gedeon, T. and Hossain, M.Z., 2024. MADE-for-ASD: A multi-atlas deep ensemble network for diagnosing Autism Spectrum Disorder. Computers in Biology and Medicine, 182, p.109083.

[18] ABIDE dataset: https://www.kaggle.com/competitions/abide.

[19] Sachdeva, Jainy, Riyaansh Mittal, Jiya Mehta, Riya Jain, and Anmol Ranjan. "Resolving autism spectrum disorder (ASD) through brain topologies using fMRI dataset with multi-layer perceptron (MLP)." Psychiatry Research: Neuroimaging 343 (2024) : 111858.

[20] Guo, Y., Liu, Y., Bakker, E.M., Guo, Y. and Lew, M.S., 2023. CNN-RNN: a large-scale hierarchical image classification framework. Multimedia tools and applications, 77(8), pp.10251 - 10271.

[21] Hendr, Amna, Umar Ozgunalp, and Meryem Erbilek Kaya. "Diagnosis of autism spectrum disorder using Recurrent Neural Networks." Electronics 12, no. 3 (2023) : 612.

Authorized licensed use limited to: BTU Cottbus-Senftenberg. Downloaded on November 08,2025 at 10:11:20 UTC from IEEE Xplore. Restrictions apply.